Flaky tests pass and fail on the same code without any change to the program under test. This study runs each test of 26 open-source Java projects one hundred times and records every outcome together with the order in which the tests ran. The share of flaky tests among the n tests is

$$\hat{p}_{\text{flaky}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\left[\min_{j} o_{i,j} \neq \max_{j} o_{i,j} \right], \tag{1}$$

and the weights of the rerun policy minimize the expected cost of a failure that stays hidden,

$$\mathbf{w}^{*} = \operatorname*{argmin}_{\mathbf{w}} \sum_{i=1}^{n} \ell\left(y_i, f(x_i; \mathbf{w})\right) + \lambda \lVert \mathbf{w} \rVert_2^2. \tag{2}$$

The detector labels 450 of 533 tests as stable, 51 as flaky, and 32 as unclear. A manual check of the flaky tests confirms 47 of the 51 labels, and the remaining four tests depend on the system clock.